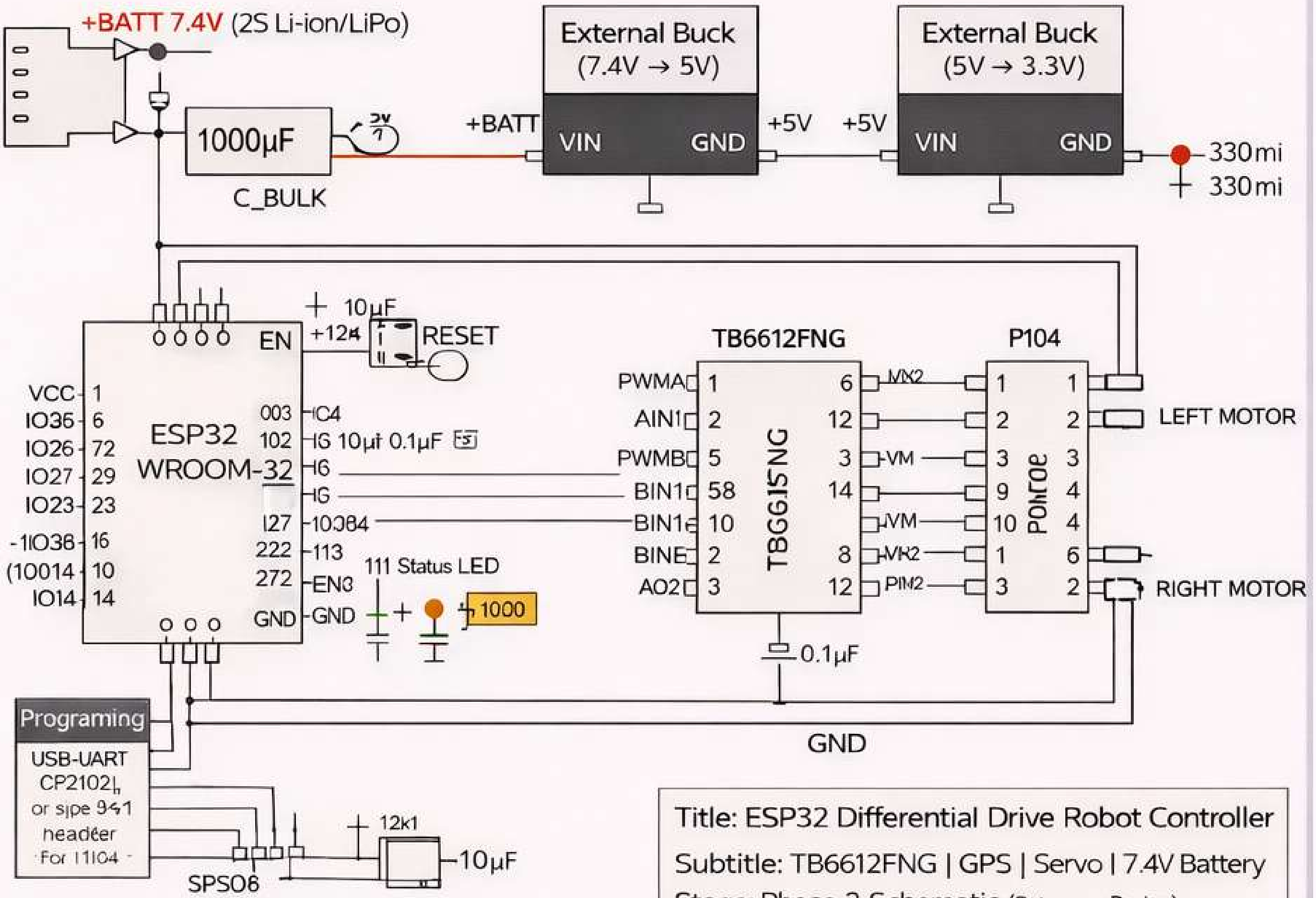
ESP32 Differential Drive Robot Controller **Schematic Structure**

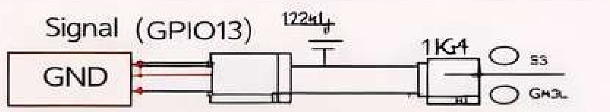


Title: ESP32 Differential Drive Robot Controller
Subtitle: TB6612FNG | GPS | Servo | 7.4V Battery
Stage: Phase 2 Schematic (Reterence Design)
⁄ CRITICAL FIX APPLIED

▪ Fixed Pin Assignments:

- PWMA → GPIO25
- AIN1 / AIN2 → GPIO26 / GPIO27
- PWMB → GPIO32
- BIN1 / BIN2 → GPIO33 / GPIO14
- Servo PWM → GPIO13
- GPS RX / TX → GPIO16 / GPIO17

⚠ Servo requires 5V, NOT 3.3V (SG30 spe: 46-0M



⚠ Servo requires 5V, NOT 3.3V (SG30 spe: 48-6M

▪ Programming: USB-UART

(CP2102) or 6-pin UART header for FTDI

• Mandatory Decoupling:

- 1000µF bulk at +BATT inout
- 10µF + 0.1µF at ESP32 VDD (+3V3)
- 10µF + 0.1µF at TB6612FNG VM (motor volege)
- 0.1µF + 0.1µF at TB6612FNG VCC (logic volege)
- 470µF near Servo +IV rail
- Status LED → GPIO2 **CRITICAL**
- EN → Pull-up + Reset button + 10µF capcitor
- GPIO0 → Pull-up + Boot button

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- External buck modules are treated as **black boxes** (VIN, GND, D2V, VOUT).
- All grounds are common. High-current motor return paths handled during PCB layout.
- Schematic-level reference design only. PCB fabrication not included.